

Mirror Object Generation

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Abstract—Generating physically correct mirror reflections remains a challenging problem for modern text-to-image diffusion models. In this work, we propose a physics-aware text-to-image generation pipeline that explicitly incorporates 3D geometric priors to enforce mirror reflection consistency. Our method automatically constructs a simple 3D scene directly from the input text prompt, procedurally generating a mirror and its physically accurate reflection. From the simulated 3D scene, a depth map is rendered and used as a conditioning signal for diffusion-based image generation. We investigate both training-based and training-free strategies for integrating these depth maps into the FLUX.1-dev backbone, including LoRA training and pretrained OminiControl conditioning. In addition, we introduce a novel geometric evaluation metric that quantitatively measures the physical correctness of mirror reflections based on the consistency of object and reflection correspondences. Experimental results demonstrate that our approach improves reflection realism and geometric correctness compared to the baseline model, while maintaining strong text-image alignment. Our implementation and evaluation code are publicly available at: <https://github.com/T2I-Mirror-Object/MirrorObjectGen>

Index Terms—Text-to-image generation, diffusion models, physics-aware image synthesis, mirror reflection, depth-conditioned Diffusion, image generation evaluation metrics

I. INTRODUCTION

Modern text-to-image (T2I) diffusion models, including multiple variants within the FLUX.1 family [1] [2] as well as Stable Diffusion family [3], [4], have demonstrated remarkable capabilities in generating photorealistic and semantically complex images. However, these models still face significant challenges when tasked with scenarios requiring strict adherence to physical laws and geometric consistency. A particularly difficult case is generating images with physically correct mirror reflections. When prompted with “a chair/cat in front of the mirror”, standard diffusion models frequently fail to produce a physically correct reflection, as shown in Figure 1.

The core limitation comes from the fact that 2D diffusion models learn statistical correlations between pixels rather than explicit physical representations. While they can mimic the appearance of a reflection, they do not inherently understand that a reflection and the real object are symmetric across the mirror plane. Recent approaches, such as MirrorFusion [5], reframe the problem as an inpainting task; however, they do not explicitly model or enforce the underlying symmetry properties.



(a) a chair in front of the mirror

(b) a cat in front of the mirror

Fig. 1. Failure cases of FLUX.1-dev Text-to-Image model when generating images containing objects and their mirror reflections.

To address this, we propose a physics-aware framework that incorporates explicit 3D priors into the generation pipeline. Our core insight is to construct a simulated 3D scene directly from the text prompt, ensuring that the object and its reflection are optically symmetric across the mirror plane.

Our method operates in three distinct stages. First, we perform semantic parsing of the prompt to identify the primary object (for example, a “chair”) and utilize Shap-E [6], a latent text-to-3D diffusion model, to “lift” the concept into a 3D mesh. Second, we explicitly construct a 3D scene using PyTorch3D [7] and Trimesh. We simulate the reflection by duplicating the object mesh and applying a geometric flip across the mirror plane. A virtual camera is positioned based on the object’s bounding box to ensure optimal visibility of both the object and its reflection.

In addition to proposing a physics-aware generation framework, we introduce a novel evaluation metric to assess the physical correctness of mirror reflections in generated images. Our metric is based on analyzing the geometric consistency between corresponding points on an object and its mirror reflection. By measuring how well these correspondences align under perspective projection, the metric provides a quantitative measurement of reflection correctness. This enables a more reliable evaluation of physics-aware image generation methods containing mirror reflections.

In the final stage, we bridge the gap between 3D geometry and photorealistic 2D generation. We render a depth map from our constructed 3D scene, which captures the spatial

relationship among all the elements in the scene. The depth map serves as a powerful condition for FLUX.1-dev [1] to generate the final image. We provide a comprehensive evaluation among two different ways of using the depth map: the first approach requires LoRA training, the second approach uses the pretrained weight from OminiControl framework [8].

In summary, our contributions are:

- We present an end-to-end, fully automatic pipeline that constructs a geometrically consistent 3D scene directly from text prompts. Then we use it as a conditioning signal for generating photorealistic images with physically accurate mirror reflections. To the best of our knowledge, we are the first to address text-to-image generation with explicit physics-aware mirror reflection modeling.
- We propose a novel geometric evaluation metric that quantitatively measures the physical correctness of mirror reflections by analyzing the consistency between objects and their reflected counterparts.
- We investigate multiple strategies for integrating depth map into the FLUX.1-dev backbone, including both training-based and training-free approaches, and provide a comprehensive evaluation of their effectiveness in generating realistic mirror reflections.

II. RELATED WORK

A. Text-to-Image Diffusion Models

The field of image synthesis has been revolutionized by denoising diffusion probabilistic models (DDPMs) [9]. Early large-scale models demonstrated unprecedented capabilities in generating high-fidelity images from natural language descriptions. More recent architectures, such as FLUX.1 [4] and Stable Diffusion 3.5 [1], have further improved image quality and prompt adherence by scaling up parameters and incorporating advanced transformer-based backbones instead of traditional U-Net backbones. While these foundation models excel at synthesizing realistic textures and lighting, they often struggle with complex spatial relationships and physical constraints [10], [11].

To address these controllability issues, several works have introduced adapter-based mechanisms, such as ControlNet [12] and T2I-Adapter [13]. These methods allow users to guide the generation process using additional spatial conditions like edge maps, depth maps, or segmentation masks. These approaches typically rely on user-provided conditions or inferred maps from existing images.

B. Text-to-3D Generation

Generating 3D content directly from text is a rapidly evolving field. Optimization-based methods, such as DreamFusion [14] and DreamGaussian [15], effectively lift 2D diffusion priors into 3D space via score distillation sampling [16], though they often suffer from slow convergence and the "Janus problem" (multi-faced objects). Feed-forward models such as Shap-E [6] and TRELLIS [17] offer significantly faster inference by learning a direct mapping from text to explicit 3D meshes. Additionally, reconstruction models like InstantMesh

[18] operate on image inputs, which requires a preceding text-to-image generation step to generate 3D meshes.

C. Mirror Reflection Synthesis

The generation of realistic mirror reflections remains a specific and challenging subset of physics-aware image synthesis. Most works attempt to solve the challenge of reconstructing mirror reflections in a 3D scene for the novel-view synthesis task [19] [20] [21]. Notably, [5] introduced a method that formulates reflection generation as a depth-conditioned inpainting problem. By masking the mirror region, the model attempts to "fill in" the reflection based on the surrounding context. Similarly, [22] proposed a large-scale dataset and training curriculum to improve the realism of these reflections. However, these inpainting-based formulations have a fundamental limitation: they treat the reflection as a completion task in 2D space rather than a geometric projection. This often results in perspective mismatches or a lack of global coherence, as the model must guess the reflection without a true understanding of the scene's 3D layout.

III. METHOD

Our hypothesis is to generate a photorealistic image where the mirror reflections are physically correct, we need an explicit 3D structure to represent the scene, then use it as a condition to the image generation process. Therefore, our method compose of 3 main stages.

A. Object Extraction and 3D Generation

Given an input prompt for the text-to-image model (for example, "a chair in front of the mirror, both the object and its mirror reflection are visible"), our pipeline first parses the text to identify and extract the description of the primary object(s). This extraction mechanism is designed to handle both single and multiple objects positioned in front of the mirror.

The extracted object text serves as the conditioning input for a pre-trained Text-to-3D model. While various architectures exist for this task, we primarily use Shap-E [6] in this work. We selected Shap-E for its ease of integration, ability to directly generate 3D meshes from text input, and low inference latency. The resulting geometry is stored in standard .obj format. Additionally, we provide an alternative implementation using TRELLIS [17] and InstantMesh [18], which requires significantly longer inference times.

B. Constructing 3D Scene

1) *Scene Composition with Mirror and Reflection:* Once the 3D meshes of primary objects are obtained from Shap-E, we construct a geometrically consistent mirrored scene with the support of Pytorch3D [7]. The goal is to create a lightweight 3D environment containing the real object, a mirror, its physically accurate reflection, and a supporting floor.

The object mesh is first grounded onto the XZ plane by translating it downward until its lowest vertex reaches $Y = 0$. To introduce viewpoint diversity across samples, a random

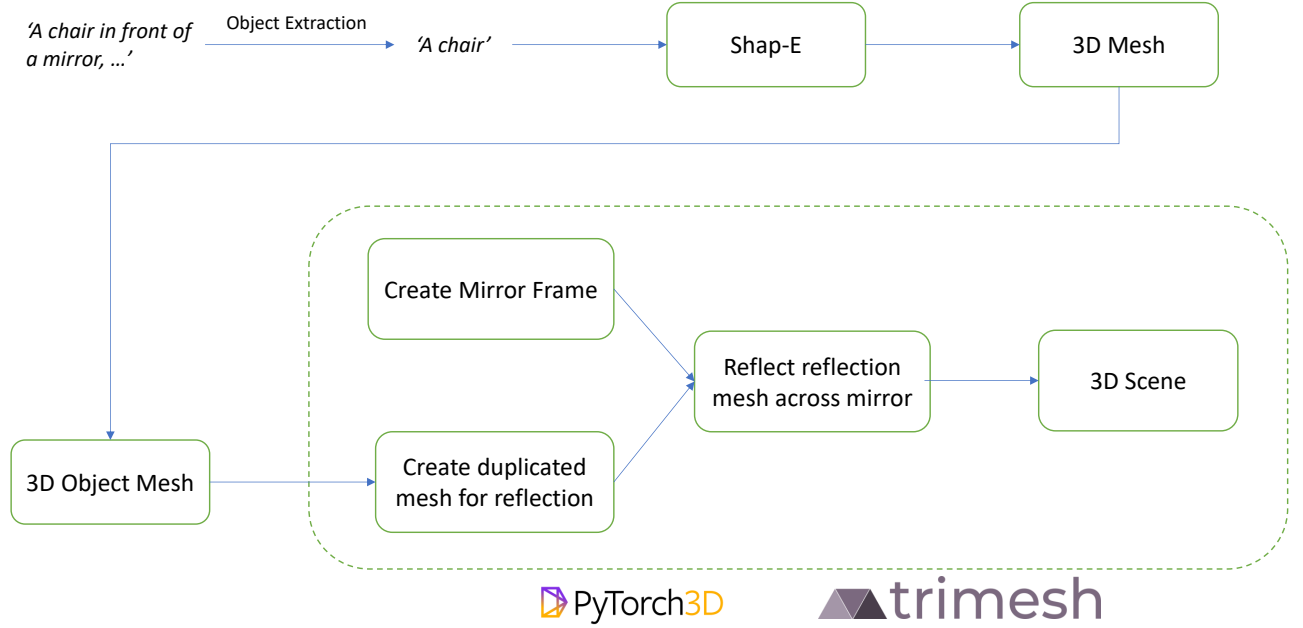


Fig. 2. The input prompt is first parsed to extract individual object descriptions (e.g., "A chair"), which are lifted into a 3D mesh using Shap-E model. We then employ PyTorch3D and Trimesh to procedurally generate a mirror frame and create a physically accurate reflection mesh, by duplicating and geometrically flipping the object mesh across the mirror frame.

rotation around the Y-axis is performed about the object’s centroid, with the angle uniformly sampled in the range $[-\pi/4, \pi/4]$.

The real object is positioned near $Z = 0$ plane. A mirror is then placed behind it at $Z = d_{gap}$, with the default value of d_{gap} being 3 units. The mirror consists of two components: (1) A rectangular frame constructed as a hollow border using four thin rectangular prisms (top, bottom, left, and right bars), (2) An inner reflective surface modeled as an extremely thin planar box with 0.01 unit in depth.

Both the frame and surface are automatically sized based on the object’s bounding box. The inner surface is precisely centered within the frame. All mirror components are implemented entirely within PyTorch3D [7] using primitive box meshes.

The mirror reflection is generated by duplicating the object mesh, applying a reflection transformation over the $Z = 0$ plane (we do this by negating all Z-coordinates), and then translating the result by $-2 \times d_{gap}$ along the Z-axis. This ensures the reflected object is positioned at the optically correct location behind the mirror plane, achieving perfect geometric symmetry.

A large, thin floor slab is added, centered at the origin and shifted downward by half its thickness so that its top surface aligns exactly with $Y = 0$.

All scene elements (object, mirror frame, mirror surface, reflection, and floor) are represented as PyTorch3D Meshes

objects without textures or materials. This untextured representation enables extremely fast scene construction (only about 1 second) while obeying geometric consistency and physics law of mirror reflections. The 3D scene will be used for deriving 2D conditioning maps in the subsequent stage.

2) *Camera Positioning*: To ensure that both the real object and its mirror reflection are fully and clearly visible in the rendered conditioning maps, we automatically determine an appropriate camera viewpoint for each scene. The camera is positioned using a perspective projection with a fixed field of view (FoV) of 60 degrees, which provides a natural perspective similar to common photographic lenses.

The camera pose is defined using spherical coordinates, starting from the positive Z-axis with a deviation of 20° – 25° in azimuth. We also employ a slightly elevated viewpoint with an elevation angle of 20° – 25° . As a result, we can balance visibility of the foreground object, the mirror plane, and the reflected object behind it.

The camera distance is computed adaptively based on the scene’s bounding box. We first estimate the extent of the visible arrangement by considering the positions of the real object (near $Z = 0$), the mirror plane (at $Z = d_{gap}$), and the reflection (at $Z = -2 \times d_{gap}$). The distance is then set such that the entire horizontal and vertical span of this arrangement subtends approximately 70–80% of the image frame, leaving moderate margins. This heuristic eliminates

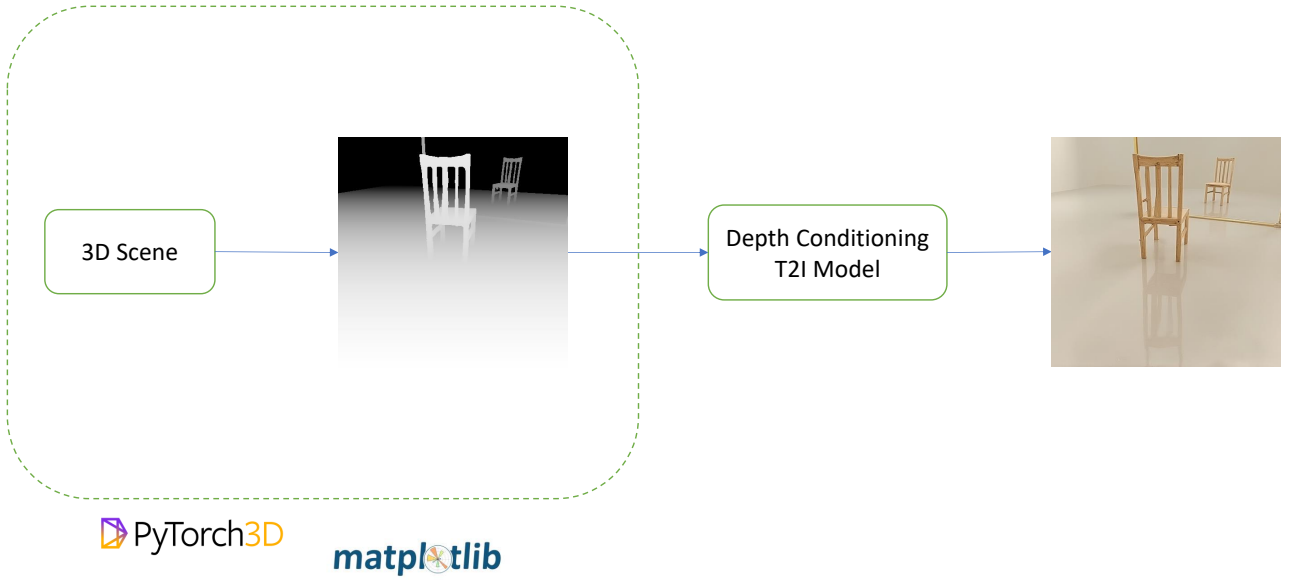


Fig. 3. We render the depth map as a 2D geometric prior from the constructed 3D scene using PyTorch3D. The depth map serve as a guidance element for the text-to-image generation process, where we evaluate multiple integration strategies including LoRA training and OminiControl framework to produce the final photorealistic image.

the need for manual camera adjustment while maximizing the informational content of the subsequent conditioning maps.

C. Depth Map Extraction

From the constructed 3D scene and the determined camera viewpoint, we derive two main conditioning maps that provide strong geometric and semantic priors to the text-to-image synthesis model: a depth map and a segmentation map. Both are extracted using PyTorch3D’s differentiable rasterizer [7] and Matplotlib [23], ensuring perfect correspondence with the underlying 3D geometry.

The depth map is rendered using a custom rasterizer that outputs normalized depth values in the camera’s normalized device coordinates. For each pixel, we retain the depth of the closest visible surface ($faces_per_pixel = 1$). Background pixels are assigned the farthest valid depth value observed in the scene, avoiding extreme or undefined values.

The raw depth buffer is then linearly normalized to the range $[0, 1]$ based on the scene’s minimum and maximum valid depths, and subsequently inverted such that closer surfaces appear brighter (near = white, far = black). This convention aligns with common practices in depth-conditioned diffusion models, which we will discuss Section III-D. The resulting grayscale image explicitly encodes the physical distance relationships critical for mirror reflections: the real object is consistently closer than the mirror plane, which in turn is closer than the reflected object.

D. Depth Conditioning Image Generation

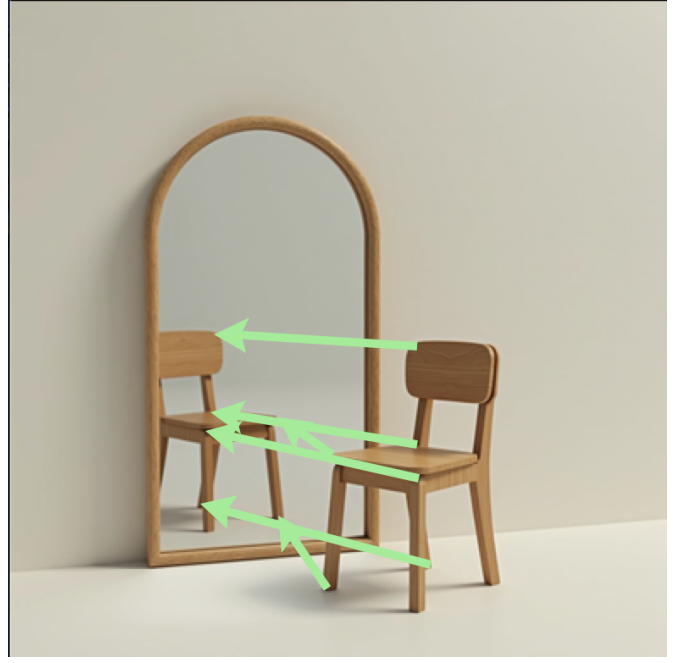
In this section, we describe how the derived depth map is integrated into text-to-image generation models to produce images with physically accurate mirror reflections. We explore two approaches: training a LoRA adapter [24] for depth-conditioned FLUX.1-dev [1], and using pretrained weight of OminiControl framework [8] with FLUX.1-dev [1]. These methods take advantage of the extracted depth map from our 3D scene construction to guide diffusion-based image synthesis, so they are likely produce symmetric and realistic mirror reflections. We provide a comprehensive comparison of their performance against baseline FLUX.1-dev model in Section V.

1) LoRA Training FLUX.1-dev with Depth Conditioning: For our first approach, we train the FLUX.1-dev model [1], a state-of-the-art diffusion transformer, using Low-Rank Adaptation (LoRA) [24] to incorporate depth map conditioning.

Dataset Preparation. We create a training dataset from the existing SynMirror dataset [5], which consists of synthetic images with ground-truth mirror reflections. From this, we extract a subset of 3,000 high-quality images featuring diverse objects in front of mirrors. For each image, we compute a corresponding depth map using Depth Anything V2 [25], a robust monocular depth estimator that provides accurate relative depths suitable for conditioning. Textual prompts are generated following a fixed template: “The {object_caption} is



(a) Real Photograph



(b) Generated Image by FLUX.1-dev

Fig. 4. Illustration of correspondence vectors connecting an object to its mirror reflection. Each vector connects a distinct feature point on the object (e.g., corner, edge) to its corresponding point in the reflection. For a physically correct reflection, these vectors should exhibit a consistent geometric relationship.

in front of the mirror, both the object and its mirror reflection are visible,” where {object_caption} is sourced directly from the SynMirror annotations (for example, “3D slanted table”). This results in triplets of (ground-truth image, depth map, prompt text). The dataset is publicly available on HuggingFace datasets at: HuggingFace Dataset

Training Procedure. We adapt the Hugging Face Diffusers [26] library’s training scripts for LoRA fine-tuning [24] on FLUX.1-dev [1]. The model is trained to minimize the denoising loss conditioned on both the text prompt and the depth map, which is concatenated as an additional input channel to the latent representation. We use a learning rate of 1×10^{-4} , batch size of 1, gradient accumulation steps of 4, LoRA rank of 64, and train for 5 epochs on a single NVIDIA A100 GPU. This approach introduces depth sensitivity without full retraining to preserve the model’s original text-to-image capabilities.

During inference, we input the text prompt alongside the depth map rendered from our 3D scene, guiding the generation toward output images with correct depth ordering.

2) *Pretrained OminiControl weight with FLUX.1-dev backbone:* Our second approach employs OminiControl [8], a minimal and universal control framework designed for Diffusion Transformer (DiT) models like FLUX.1-dev [1]. Distinct from standard Low-Rank Adaptation (LoRA) or traditional ControlNet architectures, which usually rely on simple feature addition or parameter injection, OminiControl introduces a novel mechanism for conditioning. It is driven by two key technical innovations: (1) unified sequence processing to inte-

grate condition signals as tokens directly within the attention mechanism, and (2) adaptive position encoding to assign position indices to these tokens based on task requirements to handle diverse conditions effectively.

For our pipeline, we integrate OminiControl [8] with FLUX.1-dev [1] base model to support depth conditioning: the text prompt describes the scene (“a chair in front of a mirror, both the object and its reflection are visible”), while the depth map from our 3D scene enforces geometric and spatial consistency.

We apply the depth map at a strength of 0.8 to balance fidelity to the 3D priors and creative freedom from the text.

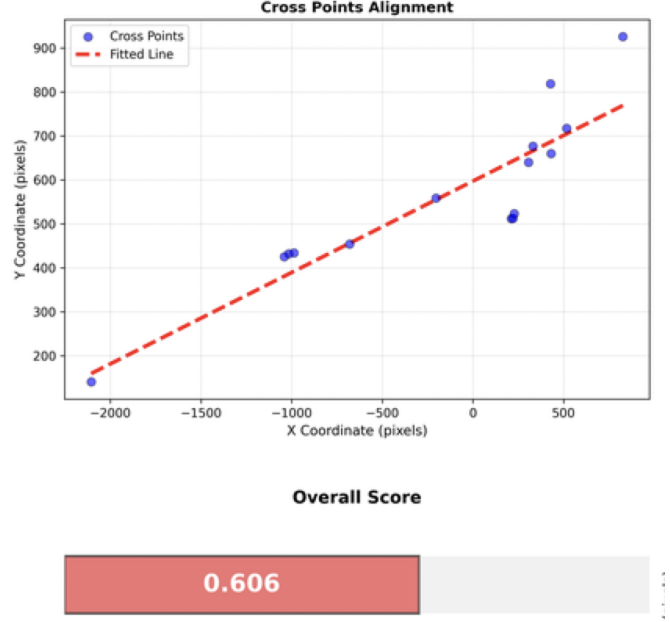
E. Novel Evaluation Metric for Physical Correctness

While existing metrics like FID [27] and CLIP score [28] are valuable for assessing general image quality and semantic alignment, they do not directly measure whether a generated mirror reflection is physically plausible. To address this gap, we propose a novel geometric metric based on empirical observations of how correspondence points behave in images with correct versus incorrect mirror reflections.

1) *Empirical Observation:* During our experiments with both real photographs and AI-generated images, we noticed an interesting pattern. When an image contains a physically correct mirror reflection, something consistent happens: if we draw lines connecting corresponding points between the real object and its reflection (for example, connecting the top-left corner of a chair to the top-left corner of its reflection), the intersection points of these lines tend to cluster together along a line in the image.



(a) Ground truth



(b) FLUX.1-dev

Fig. 5. Distribution of cross points for a physically correct reflection (left) versus an incorrect reflection (right). Cross points from correct reflections cluster tightly along a line, while incorrect reflections produce scattered cross points.

We observed this pattern across many different camera angles and scene setups. Our hypothesis is that this clustering happens because real mirror reflections maintain certain 3D geometric relationships, and these relationships show up as approximate linear patterns when the scene is captured by a camera, even with perspective distortion.

On the other hand, when baseline models such as FLUX.1-dev [1] generate physically incorrect reflections, we saw a very different pattern. The intersection points scatter randomly across the image instead of lining up. This makes intuitive sense: if the model doesn't properly understand mirror geometry, the spatial relationships between the object and reflection become inconsistent.

We developed our metric based on this observation. It measures how well these intersection points align to a straight line. We should note that this is an empirical finding rather than a theoretical guarantee, perspective projection can affect geometric relationships in complex ways. However, in practice, we found this metric works well for distinguishing correct reflections from incorrect ones.

2) *Metric Computation*: Given a generated image containing both an object and its purported mirror reflection, we compute our metric through the following procedure:

Step 1: Correspondence Vector Annotation. We identify n pairs of corresponding points between the object and its reflection, where $n \geq 6$ to ensure statistical robustness. For each pair i , we define a correspondence vector $\mathbf{v}_i$ from the object point $P_i = (x_i, y_i)$ to its reflection point $P'_i = (x'_i, y'_i)$ as shown in Figure 4

Step 2: Cross Point Calculation. For each pair of corre-

spondence vectors $\mathbf{v}_i$ and $\mathbf{v}_j$ where $i < j$, we compute their intersection point (cross point) C_{ij} by solving the system of linear equations. Given vectors defined by endpoints (P_i, P'_i) and (P_j, P'_j) , the intersection point is found by:

$$C_{ij} = P_i + t_i(P'_i - P_i) \quad (1)$$

where t_i is computed from:

$$t_i = \frac{(x_i - x_j)(y_j - y'_j) - (y_i - y_j)(x_j - x'_j)}{(x_i - x'_i)(y_j - y'_j) - (y_i - y'_i)(x_j - x'_j)} \quad (2)$$

This yields a total of $\binom{n}{2}$ cross points. For a physically correct reflection with a flat mirror, all cross points should lie approximately on a single line as shown in Figure 5.

Step 3: Line Fitting and Quality Assessment. We fit a line to the set of cross points $\{C_{ij}\}$ using ordinary least squares regression. Let the cross points be denoted as (c_k^x, c_k^y) for $k = 1, 2, \dots, N_c$, where $N_c = \binom{n}{2}$. We fit a model $y = mx + b$ by minimizing:

$$\min_{m,b} \sum_{k=1}^{N_c} (c_k^y - mc_k^x - b)^2 \quad (3)$$

To quantify the quality of this fit, we employ two complementary metrics:

1) *Coefficient of Determination (R^2)*: This metric measures how well the cross points align to the fitted line:

$$R^2 = 1 - \frac{\sum_{k=1}^{N_c} (c_k^y - \hat{c}_k^y)^2}{\sum_{k=1}^{N_c} (c_k^y - \bar{c}^y)^2} \quad (4)$$

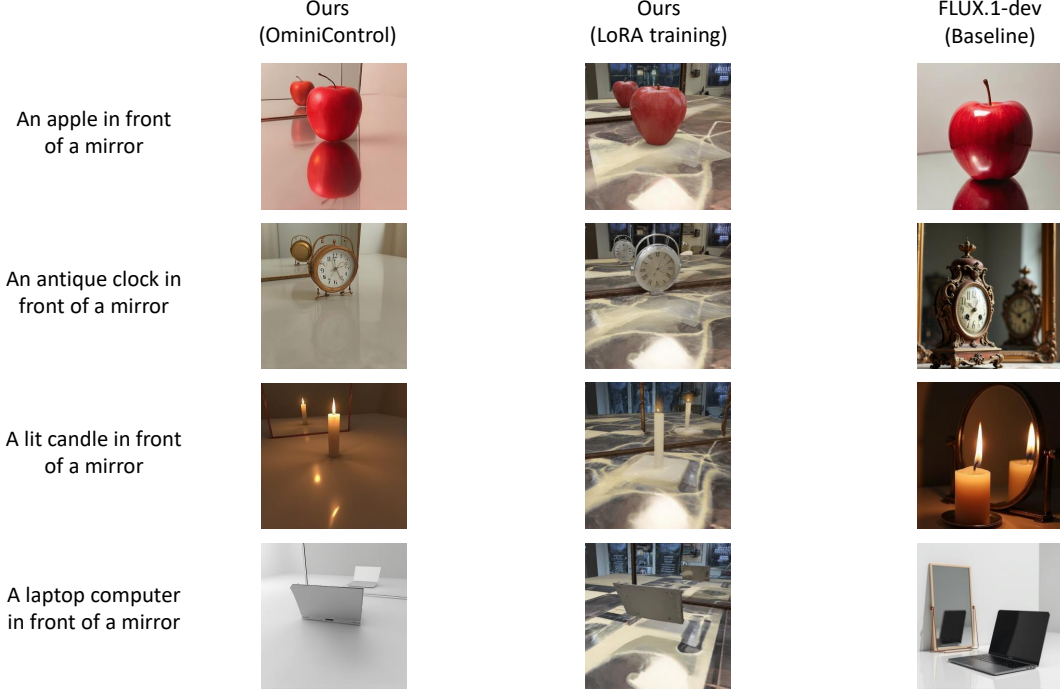


Fig. 6. **Qualitative comparisons.** We evaluated our two methods "Ours (OminiControl)" and "Ours (LoRA training)" against the FLUX.1-dev baseline model. In general, our methods generate realistic and physically correct reflections, while FLUX.1-dev struggle in some examples (missing reflections, physical violation, ...)

where $\hat{c}_k^y = mc_k^x + b$ is the predicted y -coordinate on the fitted line and $\bar{c}^y$ is the mean of all y -coordinates. Values of R^2 close to 1.0 indicate that cross points lie nearly perfectly on a line, which is characteristic of physically correct reflections.

2) *Mean Perpendicular Distance:* We also calculate the average perpendicular distance from each cross point to the fitted line. For a line in the form $y = mx + b$ (or equivalently $mx - y + b = 0$), the perpendicular distance from point (c_k^x, c_k^y) is:

$$d_k = \frac{|mc_k^x - c_k^y + b|}{\sqrt{m^2 + 1}} \quad (5)$$

The mean perpendicular distance is then:

$$\bar{d} = \frac{1}{N_c} \sum_{k=1}^{N_c} d_k \quad (6)$$

Smaller values of $\bar{d}$ indicate tighter clustering around the fitted line, suggesting higher geometric consistency.

Step 4: Normalized Score Computation. To provide a single, interpretable metric, we combine R^2 and $\bar{d}$ into a normalized score $S \in [0, 1]$ using a weighted formula:

$$S = 0.6 \cdot R^2 + 0.4 \cdot \exp\left(-\frac{\bar{d}}{\sigma}\right) \quad (7)$$

where $\sigma = 50$ pixels is a scale parameter. The R^2 term contributes 60% of the score and measures linear alignment

quality, while the distance term contributes 40% and is exponentially weighted to heavily penalize large deviations. This exponential weighting ensures that even with a high R^2 , significant outliers will noticeably reduce the overall score.

Based on our empirical observations, we categorize scores as follows: $S \geq 0.90$ indicates strong geometric consistency with physically correct reflections, $0.80 \leq S < 0.90$ indicates acceptable quality, and $S < 0.80$ suggests potential geometric inconsistencies that may indicate incorrect reflections.

3) *Practical Considerations:* Our metric offers several practical advantages for evaluating mirror reflections in generated images. First, it focuses on geometric consistency rather than perceptual similarity, providing a complementary view to traditional metrics like FID [27] or CLIP score [28]. Second, the score is interpretable: higher values correlate with better geometric alignment, which we observed to be associated with physically correct reflections in our experiments. Third, the metric is independent of appearance details like textures or colors, since it only relies on spatial point correspondences.

We should acknowledge some limitations. The metric requires manual annotation of correspondence points, which can be time-consuming. Also, while we observed that correct reflections tend to produce high scores and incorrect ones tend to produce low scores, this is an empirical finding. Factors like extreme camera angles or unusual mirror configurations might affect the metric in ways we have not fully characterized.

That said, in our experiments across various generated

TABLE I
CLIP SIMILARITY SCORE RESULTS

Models	CLIP Similarity
FLUX.1-dev	33.9525
Ours (LoRA training)	25.2411
Ours (OminiControl)	<u>33.2339</u>

images, the metric proved useful for comparing different generation methods. Images from our 3D-conditioned approach consistently achieved higher scores than baseline methods, aligning with our qualitative assessment of geometric correctness.

IV. EXPERIMENTS

Baseline. We compare our end-to-end image generation pipeline against the FLUX.1-dev [1] baseline model.

Notations. To write the name of our methods concisely, we denote **"Ours (LoRA training)"** for the approach that performs LoRA training to enable the original FLUX.1-dev model to condition on both prompt text and depth map during generation process. Meanwhile, **"Ours (OminiControl)"** will be used to indicate using pretrained OminiControl weight with FLUX.1-dev backbone, also for generating final image based on prompt text and depth map.

Metrics. We evaluate our method and the baseline method using primarily the novel metric that we propose in Section III-E. We also provide quantitative results using CLIP [28] Similarity Score to evaluate text-image consistency between the generated image and the prompt text.

Benchmarks. We have created 115 prompts with diverse objects following the prompt template: "A/An object_name in front of a mirror, both the object and its mirror reflection are visible.". The full inference images on this benchmark for both our methods and the baseline model are available at: Drive

V. RESULTS

A. Qualitative Results

Figure 6 show some examples of prompts and the images generated by both our methods and the baseline FLUX.1-dev model [1]. In general, our methods successfully generate physics-aware and realistic mirror reflections, while the baseline model fail in most examples.

B. CLIP Similarity Results

Table I shows the Average CLIP Similarity Score for both our methods and the baseline FLUX.1-dev [1], using the benchmark prompt dataset. The best result is shown in bold, while the second best is underlined. Our method that uses the OminiControl weight still achieves good CLIP similarity score like the baseline model, while ensuring the physical correctness of mirror reflections as shown in Qualitative Results section.

VI. CONCLUSION

In this work, we addressed the challenge of generating physically correct mirror reflections in text-to-image diffusion models by introducing a physics-aware generation pipeline with explicit 3D scene modeling. By automatically constructing a simulated 3D scene from the input prompt and using depth maps as conditioning signals, our approach enforces physics rule of symmetry between objects and their mirror reflections, which overcomes the limitation of original 2D diffusion models. We demonstrated that both training-based and training-free integration strategies can effectively use these 3D priors to produce more realistic and physically plausible results.

In addition, we proposed a novel geometric evaluation metric that quantitatively measures the physical correctness of mirror reflections. Experimental results show that our 3D-conditioned methods consistently outperform baseline models in terms of reflection accuracy while maintaining strong text-image alignment.

There are several promising directions for future work. First, the 3D scene construction stage can be further optimized to improve object shape quality and diversity, including more accurate object placement and mirror configurations. Second, beyond depth maps, future work could explore additional conditioning signals such as segmentation maps, edge maps or surface normals to guide text-to-image diffusion models. Finally, building larger and more diverse benchmarks for mirror reflection scenarios would enable more comprehensive and reliable evaluation of physics-aware image generation methods.

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